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# Questions

## 1. Are the calculations in some of the reports incorrect?

- Yes. All three reports seem to have some errors but the calculations are mostly ok.
- The errors are in the figures especially to compute the cashflow.

| Report | Error | Corrected NPV |
|---|---|---|
| Andreu (EOQ) | • Andreu used costs that include fixed costs which don't change with the size of Q: $K$ = 776,435, $c$ = 2,254<br>• Depreciation (689,523) and overhead (712/kg) are fixed and should be excluded.<br>• Higher $K$ increases $Q^*$, while higher $c$ descreases it (they cancel near 3,000 kg) | $ -75,182 |
| Paul (NPV) | • Paul factors the one-time inventory release every year (1,963,985/yr) which is wrong.<br>• The holding cost used (17%) already includes the 12% cost of capital, and he also counts the inventory cash flow (cost of capital counted twice)<br>• Inconsistent inventory figures (1,971,448 vs 1,963,985) | $ -403,460 |
| Pierre (EVA) | • EVA setup is correct<br>• NOPAT uses the 17% holding cost, which already includes 12%<br>• The capital charge adds 12% again (effective rate of 29%) | $ 1,021,151 |

- Paul and Pierre do correctly exclude fixed costs from $K$ and $c$

## 2. If the calculations were correct, which approach is right?

- All three, if they:
  - use only incremental costs
  - charge the cost of capital exactly once
- Then all give 1,247 kg and $ 1,118,856

$$\mathrm{NPV}(q) = \mathrm{MVA}(q) = \frac{\mathrm{TRC}(q_0) - \mathrm{TRC}(q)}{r}$$

- Why they agree:
  - **NPV = MVA:** PV of all future EVAs = NPV, if the capital charge is on the same capital the cash flows create
  - **NPV = EOQ:** in a perpetuity, $r$ × one-time inventory release = annual cost of capital on that inventory (max NPV = min annual cost with $h = h_{\mathrm{mat}} + r$)
- Correct objective: maximise value (NPV)
- EOQ minimises cost, which is equivalent here because revenue does not depend on batch size

## 3. Is one approach always superior to the others?

- No, but they do not cover the same ground

| Method | Strengths | Limits |
|---|---|---|
| EOQ | • Simple closed-form formula<br>• Needs only four inputs<br>• Shows the trade-off clearly | • Only valid under its assumptions (Q4)<br>• Cost of capital must be built into $h$ correctly |
| NPV | • Most general<br>• Handles taxes, finite horizons, changing demand, transition timing, one-time costs | • Needs a full cash-flow model<br>• Easy to get working capital wrong (Paul) |
| EVA / MVA | • Equals NPV if done consistently<br>• Links to managers' performance metrics | • Mixes book and market values<br>• One year's EVA can mislead; only MVA is a decision rule<br>• Easy to double-count capital cost (Pierre) |

- NPV is the **reference**: EVA must match it, and EOQ is a special case of it
- Most general ≠ best tool for routine decisions

## 4. Are the other approaches inappropriate? What assumptions would invalidate them?

- None is inappropriate for Mainclas
- The case meets EOQ's assumptions: stable demand, one plant with spare capacity, no taxes
- EOQ stops being exact when:
  - **Demand varies or is uncertain**: safety stock, Wagner-Whitin, or simulation
  - **Horizon is finite** (end of life, patent expiry, 2-3-year shelf life) (no perpetuity; coverage must stay below shelf life)
  - **Capacity is shared**: setup cost includes lost production (shutdowns of "days or weeks") so, joint scheduling (ELSP)
  - **Production is not instant**: use EPQ: average inventory $= q(1 - d/p)/2$, not $q/2$
  - **Taxes exist**: use after-tax costs and after-tax WACC; simple EOQ no longer matches NPV
- NPV and EVA can model all of these (they fail only when misapplied)

## 5. Are some approaches just rough approximations?

- Yes, even the corrected models simplify reality:
  - **EOQ uses averages:** an exact model discounts each setup and each unit of inventory at its actual date so the optimum shifts slightly
  - **Inventory release is not instant:** stock falls only as the current 3,000-kg campaign is used up (the release comes over year 1, so the real NPV is slightly lower)
  - **Lumpy flows are replaced by averages** (Paul's own stated assumption)
- This matters little because the value curve is flat near the optimum:

| $q$ (kg) | 800 | 1,000 | 1,100 | 1,247 | 1,400 | 1,600 | 2,000 |
|---|---|---|---|---|---|---|---|
| **NPV ($)** | 846,078 | 1,052,230 | 1,097,405 | 1,118,856 | 1,100,588 | 1,033,782 | 809,182 |
| **% of max** | 76% | 94% | 98% | 100% | 98% | 92% | 72% |

- 1,000–1,600 kg (more than 90% of max value)
- Getting the method right matters more than precision:
  - Paul's error: a $ 1,118,856 gain becomes a 403,460 loss
  - Rounding to 1,200 kg: still 100% of max value

**Bottom line:**

- The reports disagree because of modelling errors, not because the methods disagree
- Done correctly, EOQ, NPV and EVA give the same answer
- Recommendation: cut the campaign to about 1,247 kg